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MySQL Assignment 2 – Querying Data

1. Distinct Values:

a) A query to retrieve distinct salaries from the Employees table.

```
/* query to retrieve distinct salaries from the Employees table.*/

SELECT DISTINCT salary FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

Employees 11

2. Alias (AS):

a) Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

```
/*Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively. */

SELECT age AS Employee_Age, salary AS Employee_Salary FROM Employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Employee_Age	Employee_Salary
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00
29	80000.00
28	60000.00

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3. Where Clause & Operators:

a) Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

```sql
/*Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01. */

SELECT * FROM Employees WHERE salary>50000 AND hire_date<'2016-01-01';
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

b) Find the employee whose designation is missing and fill it with "Data Scientist".

```sql
/*the employee whose designation is missing*/

SELECT * FROM Employees WHERE designation IS NULL;
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5004	Kiara Malhotra	F	29	2015-07-05	NULL	8	3	70000.00

Employee whose designation is missing

```sql
/*the employee whose designation is missing and fill it with "Data Scientist". */
SELECT * FROM Employees WHERE designation IS NULL;

UPDATE employees SET designation ='Data Scientist' WHERE designation IS NULL;
SELECT * FROM Employees WHERE designation = 'Data Scientist';
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5004	Kiara Malhotra	F	29	2015-07-05	Data Scientist	8	3	70000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00

Missing designation updated successfully to Data Scientist

Sorting and Grouping Data:

1. ORDER BY:

a) Find employees sorted by department ID in ascending order and salary in descending order.

```sql
/*Find employees sorted by department ID in ascending order and salary in descending order. */

SELECT * FROM Employees ORDER BY department_id ASC, salary DESC;
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	3	2	75000.00
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	5	1	95000.00

Employees 1

2. LIMIT:

a) Display the first 5 employees hired in the year 2018.

```sql
/* Display the first 5 employees hired in the year 2018.*/

SELECT * FROM Employees WHERE YEAR(hire_date)=2018 ORDER BY hire_date ASC LIMIT 5;
```

employee_id	employee_name	gender	age	hire_date	designation	department_id	location_id	salary
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Aggregate Functions:

a) Calculate the sum of all salaries in the Finance department.

The department_id of Finance department is 7

```sql
/* Calculate the sum of all salaries in the Finance department. */

SELECT SUM(salary) AS Total_Finance_salary FROM Employees WHERE department_id =7;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

Total_Finance_salary
170000.00

b) Find the minimum age among all employees.

```sql
/* Find the minimum age among all employees. */

SELECT MIN(age) AS Minimum_age FROM Employees;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

Minimum_age
25

4. GROUP BY:

a) List the maximum salary for each location.

```sql
/* List the maximum salary for each location. */

SELECT e.location_id,l.location_name,MAX(salary) AS Maximum_salary FROM Employees e LEFT JOIN Location l ON e.location_id=l.location_id GROUP BY location_id;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

location_id	location_name	Maximum_salary
4	Pune	100000.00
1	Chennai	95000.00
2	Bangalore	95000.00
3	Hyderabad	90000.00

b) Calculate the average salary for each designation containing the word 'Analyst'.

```sql
/* Calculate the average salary for each designation containing the word 'Analyst'. */

SELECT designation,AVG(salary) AS Avg_salary FROM Employees WHERE designation LIKE '%Analyst%' GROUP BY designation;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

designation	Avg_salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

5. HAVING:

a) Find departments with less than 3 employees.

```sql
/* Find departments with less than 3 employees. */

SELECT e.department_id,d.department_name, count(*) AS Employee_count FROM Employees e
LEFT JOIN Departments d ON e.department_id=d.department_id
GROUP BY department_id HAVING count(*)<3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	department_name	Employee_count
1	Software Development	1
5	Product Management	2
6	Content Creation	2
7	Finance	2
9	Research and Development	1
11	Business Development	2
13	Operations	1

b) Find locations with female employees whose average age is below 30.

```sql
/* Find locations with female employees whose average age is below 30.*/

SELECT e.location_id,l.location_name,ROUND(avg(age)) AS Avg_age FROM Employees e
LEFT JOIN Location l ON e.location_id=l.location_id WHERE gender='F'
GROUP BY location_id HAVING Avg_age<30;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	location_name	Avg_age
2	Bangalore	27
1	Chennai	27
4	Pune	29

Joins:

1. Inner Join:

a) List employee names, their designations, and department names where employees are assigned to a department.

```sql
/* List employee names, their designations, and department names where employees are assigned to a department.*/

SELECT e.employee_name,e.designation,d.department_name FROM Employees e
INNER JOIN Departments d ON e.department_id=d.department_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	designation	department_name
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science
Avni Iyengar	Data Scientist	Data Science
Vivaan Singh	Business Analyst	Data Science
Kiara Malhotra	Data Scientist	Design
Dhruv Shetty	UI Developer	Design
Diya Jha	Graphic Designer	Design
Ishaan Kumar	Financial Analyst	Finance
Advik Kapoor	Finance Analyst	Finance
Zoya Khan	Legal Counsel	Human Resources
Ishika Patel	Talent Acquisition Specialist	Human Resources
Aadhya Iyengar	HR Specialist	Human Resources
Reyansh Singh	Network Engineer	IT
Atharv Yadav	Systems Administrator	IT
Kabir Nair	IT Support Specialist	IT
Arjun Kumar	Quality Assurance Analyst	IT
Anushka Singh	Marketing Manager	Marketing
Saanvi Patel	Marketing Analyst	Marketing

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2. Left Join:

a) List all departments along with the total number of employees in each department, including departments with no employees.

```sql
/* List all departments along with the total number of employees in each department, including departments with no employees.*/

SELECT d.department_id,d.department_name,count(*) AS Employee_count FROM Departments d
LEFT JOIN Employees e ON d.department_id=e.department_id
GROUP BY department_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	department_name	Employee_count
1	Software Development	1
2	Marketing	3
3	Data Science	3
4	Human Resources	3
5	Product Management	2
6	Content Creation	2
7	Finance	2
8	Design	3
9	Research and Development	1
10	Customer Support	3
11	Business Development	2
12	IT	4
13	Operations	1

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3. Right Join:

a) Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```
/* Display all locations along with the names of employees assigned to each locatio

SELECT l.location_name,e.employee_name FROM Employees e
RIGHT JOIN Location l ON e.location_id=l.location_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_name	employee_name
Chennai	Anvi Chaudhary
Chennai	Saanvi Patel
Chennai	Ishaan Kumar
Chennai	Aarav Nair
Chennai	Aryan Shetty
Chennai	Anaya Kapoor
Chennai	Sara Iyer
Hyderabad	Kiara Malhotra
Hyderabad	Anushka Singh
Hyderabad	Kiaan Desai
Hyderabad	Arnav Rao
Hyderabad	Ishan Mishra
Hyderabad	Advik Kapoor
Hyderabad	Ananya Paul
Pune	Vihaan Singh
Pune	Diya Jha
Pune	Atharv Yadav
Pune	Zoya Khan
Pune	Ishika Patel
Pune	Aadhya Iyengar
Pune	Avni Iyengar